

100 Integrals

Full document export regression: problems first, complete answer key last.

PROBLEM-001

$$\int_0^{\infty} x^1 e^{-2x} dx$$

PROBLEM-002

$$\int_0^{\infty} x^2 e^{-2x} dx$$

PROBLEM-003

$$\int_0^{\infty} x^3 e^{-2x} dx$$

PROBLEM-004

$$\int_0^{\infty} x^4 e^{-2x} dx$$

PROBLEM-005

$$\int_0^{\infty} x^5 e^{-2x} dx$$

PROBLEM-006

$$\int_0^{\infty} x^6 e^{-2x} dx$$

PROBLEM-007

$$\int_0^{\infty} x^7 e^{-2x} dx$$

PROBLEM-008

$$\int_0^{\infty} x^8 e^{-2x} dx$$

PROBLEM-009

$$\int_0^{\infty} x^9 e^{-2x} dx$$

PROBLEM-010

$$\int_0^{\infty} x^{10} e^{-2x} dx$$

PROBLEM-011

$$\int_0^{\infty} x^{11} e^{-2x} dx$$

PROBLEM-012

$$\int_0^{\infty} x^{12} e^{-2x} dx$$

PROBLEM-013

$$\int_0^{\infty} x^{13} e^{-2x} dx$$

PROBLEM-014

$$\int_0^{\infty} x^{14} e^{-2x} dx$$

PROBLEM-015

$$\int_0^{\infty} x^{15} e^{-2x} dx$$

PROBLEM-016

$$\int_0^{\infty} x^{16} e^{-2x} dx$$

PROBLEM-017

$$\int_0^{\infty} x^{17} e^{-2x} dx$$

PROBLEM-018

$$\int_0^{\infty} x^{18} e^{-2x} dx$$

PROBLEM-019

$$\int_0^{\infty} x^{19} e^{-2x} dx$$

PROBLEM-020

$$\int_0^{\infty} x^{20} e^{-2x} dx$$

PROBLEM-021

$$\int_0^{\infty} x^{21} e^{-2x} dx$$

PROBLEM-022

$$\int_0^{\infty} x^{22} e^{-2x} dx$$

PROBLEM-023

$$\int_0^{\infty} x^{23} e^{-2x} dx$$

PROBLEM-024

$$\int_0^{\infty} x^{24} e^{-2x} dx$$

PROBLEM-025

$$\int_0^{\infty} x^{25} e^{-2x} dx$$

PROBLEM-026

$$\int_0^{\infty} x^{26} e^{-2x} dx$$

PROBLEM-027

$$\int_0^{\infty} x^{27} e^{-2x} dx$$

PROBLEM-028

$$\int_0^{\infty} x^{28} e^{-2x} dx$$

PROBLEM-029

$$\int_0^{\infty} x^{29} e^{-2x} dx$$

PROBLEM-030

$$\int_0^{\infty} x^{30} e^{-2x} dx$$

PROBLEM-031

$$\int_0^{\infty} x^{31} e^{-2x} dx$$

PROBLEM-032

$$\int_0^{\infty} x^{32} e^{-2x} dx$$

PROBLEM-033

$$\int_0^{\infty} x^{33} e^{-2x} dx$$

PROBLEM-034

$$\int_0^{\infty} x^{34} e^{-2x} dx$$

PROBLEM-035

$$\int_0^{\infty} x^{35} e^{-2x} dx$$

PROBLEM-036

$$\int_0^{\infty} x^{36} e^{-2x} dx$$

PROBLEM-037

$$\int_0^{\infty} x^{37} e^{-2x} dx$$

PROBLEM-038

$$\int_0^{\infty} x^{38} e^{-2x} dx$$

PROBLEM-039

$$\int_0^{\infty} x^{39} e^{-2x} dx$$

PROBLEM-040

$$\int_0^{\infty} x^{40} e^{-2x} dx$$

PROBLEM-041

$$\int_0^{\infty} x^{41} e^{-2x} dx$$

PROBLEM-042

$$\int_0^{\infty} x^{42} e^{-2x} dx$$

PROBLEM-043

$$\int_0^{\infty} x^{43} e^{-2x} dx$$

PROBLEM-044

$$\int_0^{\infty} x^{44} e^{-2x} dx$$

PROBLEM-045

$$\int_0^{\infty} x^{45} e^{-2x} dx$$

PROBLEM-046

$$\int_0^{\infty} x^{46} e^{-2x} dx$$

PROBLEM-047

$$\int_0^{\infty} x^{47} e^{-2x} dx$$

PROBLEM-048

$$\int_0^{\infty} x^{48} e^{-2x} dx$$

PROBLEM-049

$$\int_0^{\infty} x^{49} e^{-2x} dx$$

PROBLEM-050

$$\int_0^{\infty} x^{50} e^{-2x} dx$$

PROBLEM-051

$$\int_0^{\infty} x^{51} e^{-2x} dx$$

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$$\int_0^{\infty} x^{52} e^{-2x} dx$$

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$$\int_0^{\infty} x^{53} e^{-2x} dx$$

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$$\int_0^{\infty} x^{55} e^{-2x} dx$$

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$$\int_0^{\infty} x^{56} e^{-2x} dx$$

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$$\int_0^{\infty} x^{57} e^{-2x} dx$$

PROBLEM-058

$$\int_0^{\infty} x^{58} e^{-2x} dx$$

PROBLEM-059

$$\int_0^{\infty} x^{59} e^{-2x} dx$$

PROBLEM-060

$$\int_0^{\infty} x^{60} e^{-2x} dx$$

PROBLEM-061

$$\int_0^{\infty} x^{61} e^{-2x} dx$$

PROBLEM-062

$$\int_0^{\infty} x^{62} e^{-2x} dx$$

PROBLEM-063

$$\int_0^{\infty} x^{63} e^{-2x} dx$$

PROBLEM-064

$$\int_0^{\infty} x^{64} e^{-2x} dx$$

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$$\int_0^{\infty} x^{65} e^{-2x} dx$$

PROBLEM-066

$$\int_0^{\infty} x^{66} e^{-2x} dx$$

PROBLEM-067

$$\int_0^{\infty} x^{67} e^{-2x} dx$$

PROBLEM-068

$$\int_0^{\infty} x^{68} e^{-2x} dx$$

PROBLEM-069

$$\int_0^{\infty} x^{69} e^{-2x} dx$$

PROBLEM-070

$$\int_0^{\infty} x^{70} e^{-2x} dx$$

PROBLEM-071

$$\int_0^{\infty} x^{71} e^{-2x} dx$$

PROBLEM-072

$$\int_0^{\infty} x^{72} e^{-2x} dx$$

PROBLEM-073

$$\int_0^{\infty} x^{73} e^{-2x} dx$$

PROBLEM-074

$$\int_0^{\infty} x^{74} e^{-2x} dx$$

PROBLEM-075

$$\int_0^{\infty} x^{75} e^{-2x} dx$$

PROBLEM-076

$$\int_0^{\infty} x^{76} e^{-2x} dx$$

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$$\int_0^{\infty} x^{77} e^{-2x} dx$$

PROBLEM-078

$$\int_0^{\infty} x^{78} e^{-2x} dx$$

PROBLEM-079

$$\int_0^{\infty} x^{79} e^{-2x} dx$$

PROBLEM-080

$$\int_0^{\infty} x^{80} e^{-2x} dx$$

PROBLEM-081

$$\int_0^{\infty} x^{81} e^{-2x} dx$$

PROBLEM-082

$$\int_0^{\infty} x^{82} e^{-2x} dx$$

PROBLEM-083

$$\int_0^{\infty} x^{83} e^{-2x} dx$$

PROBLEM-084

$$\int_0^{\infty} x^{84} e^{-2x} dx$$

PROBLEM-085

$$\int_0^{\infty} x^{85} e^{-2x} dx$$

PROBLEM-086

$$\int_0^{\infty} x^{86} e^{-2x} dx$$

PROBLEM-087

$$\int_0^{\infty} x^{87} e^{-2x} dx$$

PROBLEM-088

$$\int_0^{\infty} x^{88} e^{-2x} dx$$

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$$\int_0^{\infty} x^{89} e^{-2x} dx$$

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$$\int_0^{\infty} x^{92} e^{-2x} dx$$

PROBLEM-093

$$\int_0^{\infty} x^{93} e^{-2x} dx$$

PROBLEM-094

$$\int_0^{\infty} x^{94} e^{-2x} dx$$

PROBLEM-095

$$\int_0^{\infty} x^{95} e^{-2x} dx$$

PROBLEM-096

$$\int_0^{\infty} x^{96} e^{-2x} dx$$

PROBLEM-097

$$\int_0^{\infty} x^{97} e^{-2x} dx$$

PROBLEM-098

$$\int_0^{\infty} x^{98} e^{-2x} dx$$

PROBLEM-099

$$\int_0^{\infty} x^{99} e^{-2x} dx$$

PROBLEM-100

$$\int_0^{\infty} x^{100} e^{-2x} dx$$

Solutions

SOLUTION-001

Using the Gamma integral with the substitution $t = 2x$,
$$\int_0^{\infty} x^1 e^{-2x} dx = \frac{\Gamma(2)}{2^2}.$$

SOLUTION-002

Using the Gamma integral with the substitution $t = 2x$,
$$\int_0^{\infty} x^2 e^{-2x} dx = \frac{\Gamma(3)}{2^3}.$$

SOLUTION-003

Using the Gamma integral with the substitution $t = 2x$,
$$\int_0^{\infty} x^3 e^{-2x} dx = \frac{\Gamma(4)}{2^4}.$$

SOLUTION-004

Using the Gamma integral with the substitution $t = 2x$,
$$\int_0^{\infty} x^4 e^{-2x} dx = \frac{\Gamma(5)}{2^5}.$$

SOLUTION-005

Using the Gamma integral with the substitution $t = 2x$,
$$\int_0^{\infty} x^5 e^{-2x} dx = \frac{\Gamma(6)}{2^6}.$$

SOLUTION-006

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^6 e^{-2x} dx = \frac{\Gamma(7)}{2^7}.$$

SOLUTION-007

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^7 e^{-2x} dx = \frac{\Gamma(8)}{2^8}.$$

SOLUTION-008

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^8 e^{-2x} dx = \frac{\Gamma(9)}{2^9}.$$

SOLUTION-009

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^9 e^{-2x} dx = \frac{\Gamma(10)}{2^{10}}.$$

SOLUTION-010

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{10} e^{-2x} dx = \frac{\Gamma(11)}{2^{11}}.$$

SOLUTION-011

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{11} e^{-2x} dx = \frac{\Gamma(12)}{2^{12}}.$$

SOLUTION-012

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{12} e^{-2x} dx = \frac{\Gamma(13)}{2^{13}}.$$

SOLUTION-013

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{13} e^{-2x} dx = \frac{\Gamma(14)}{2^{14}}.$$

SOLUTION-014

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{14} e^{-2x} dx = \frac{\Gamma(15)}{2^{15}}.$$

SOLUTION-015

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{15} e^{-2x} dx = \frac{\Gamma(16)}{2^{16}}.$$

SOLUTION-016

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{16} e^{-2x} dx = \frac{\Gamma(17)}{2^{17}}.$$

SOLUTION-017

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{17} e^{-2x} dx = \frac{\Gamma(18)}{2^{18}}.$$

SOLUTION-018

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{18} e^{-2x} dx = \frac{\Gamma(19)}{2^{19}}.$$

SOLUTION-019

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{19} e^{-2x} dx = \frac{\Gamma(20)}{2^{20}}.$$

SOLUTION-020

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{20} e^{-2x} dx = \frac{\Gamma(21)}{2^{21}}.$$

SOLUTION-021

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{21} e^{-2x} dx = \frac{\Gamma(22)}{2^{22}}.$$

SOLUTION-022

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{22} e^{-2x} dx = \frac{\Gamma(23)}{2^{23}}.$$

SOLUTION-023

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{23} e^{-2x} dx = \frac{\Gamma(24)}{2^{24}}.$$

SOLUTION-024

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{24} e^{-2x} dx = \frac{\Gamma(25)}{2^{25}}.$$

SOLUTION-025

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{25} e^{-2x} dx = \frac{\Gamma(26)}{2^{26}}.$$

SOLUTION-026

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{26} e^{-2x} dx = \frac{\Gamma(27)}{2^{27}}.$$

SOLUTION-027

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{27} e^{-2x} dx = \frac{\Gamma(28)}{2^{28}}.$$

SOLUTION-028

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{28} e^{-2x} dx = \frac{\Gamma(29)}{2^{29}}.$$

SOLUTION-029

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{29} e^{-2x} dx = \frac{\Gamma(30)}{2^{30}}.$$

SOLUTION-030

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{30} e^{-2x} dx = \frac{\Gamma(31)}{2^{31}}.$$

SOLUTION-031

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{31} e^{-2x} dx = \frac{\Gamma(32)}{2^{32}}.$$

SOLUTION-032

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{32} e^{-2x} dx = \frac{\Gamma(33)}{2^{33}}.$$

SOLUTION-033

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{33} e^{-2x} dx = \frac{\Gamma(34)}{2^{34}}.$$

SOLUTION-034

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{34} e^{-2x} dx = \frac{\Gamma(35)}{2^{35}}.$$

SOLUTION-035

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{35} e^{-2x} dx = \frac{\Gamma(36)}{2^{36}}.$$

SOLUTION-036

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{36} e^{-2x} dx = \frac{\Gamma(37)}{2^{37}}.$$

SOLUTION-037

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{37} e^{-2x} dx = \frac{\Gamma(38)}{2^{38}}.$$

SOLUTION-038

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{38} e^{-2x} dx = \frac{\Gamma(39)}{2^{39}}.$$

SOLUTION-039

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{39} e^{-2x} dx = \frac{\Gamma(40)}{2^{40}}.$$

SOLUTION-040

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{40} e^{-2x} dx = \frac{\Gamma(41)}{2^{41}}.$$

SOLUTION-041

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{41} e^{-2x} dx = \frac{\Gamma(42)}{2^{42}}.$$

SOLUTION-042

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{42} e^{-2x} dx = \frac{\Gamma(43)}{2^{43}}.$$

SOLUTION-043

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{43} e^{-2x} dx = \frac{\Gamma(44)}{2^{44}}.$$

SOLUTION-044

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{44} e^{-2x} dx = \frac{\Gamma(45)}{2^{45}}.$$

SOLUTION-045

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{45} e^{-2x} dx = \frac{\Gamma(46)}{2^{46}}.$$

SOLUTION-046

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{46} e^{-2x} dx = \frac{\Gamma(47)}{2^{47}}.$$

SOLUTION-047

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{47} e^{-2x} dx = \frac{\Gamma(48)}{2^{48}}.$$

SOLUTION-048

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{48} e^{-2x} dx = \frac{\Gamma(49)}{2^{49}}.$$

SOLUTION-049

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{49} e^{-2x} dx = \frac{\Gamma(50)}{2^{50}}.$$

SOLUTION-050

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{50} e^{-2x} dx = \frac{\Gamma(51)}{2^{51}}.$$

SOLUTION-051

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{51} e^{-2x} dx = \frac{\Gamma(52)}{2^{52}}.$$

SOLUTION-052

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{52} e^{-2x} dx = \frac{\Gamma(53)}{2^{53}}.$$

SOLUTION-053

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{53} e^{-2x} dx = \frac{\Gamma(54)}{2^{54}}.$$

SOLUTION-054

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{54} e^{-2x} dx = \frac{\Gamma(55)}{2^{55}}.$$

SOLUTION-055

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{55} e^{-2x} dx = \frac{\Gamma(56)}{2^{56}}.$$

SOLUTION-056

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{56} e^{-2x} dx = \frac{\Gamma(57)}{2^{57}}.$$

SOLUTION-057

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{57} e^{-2x} dx = \frac{\Gamma(58)}{2^{58}}.$$

SOLUTION-058

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{58} e^{-2x} dx = \frac{\Gamma(59)}{2^{59}}.$$

SOLUTION-059

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{59} e^{-2x} dx = \frac{\Gamma(60)}{2^{60}}.$$

SOLUTION-060

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{60} e^{-2x} dx = \frac{\Gamma(61)}{2^{61}}.$$

SOLUTION-061

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{61} e^{-2x} dx = \frac{\Gamma(62)}{2^{62}}.$$

SOLUTION-062

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{62} e^{-2x} dx = \frac{\Gamma(63)}{2^{63}}.$$

SOLUTION-063

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{63} e^{-2x} dx = \frac{\Gamma(64)}{2^{64}}.$$

SOLUTION-064

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{64} e^{-2x} dx = \frac{\Gamma(65)}{2^{65}}.$$

SOLUTION-065

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{65} e^{-2x} dx = \frac{\Gamma(66)}{2^{66}}.$$

SOLUTION-066

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{66} e^{-2x} dx = \frac{\Gamma(67)}{2^{67}}.$$

SOLUTION-067

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{67} e^{-2x} dx = \frac{\Gamma(68)}{2^{68}}.$$

SOLUTION-068

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{68} e^{-2x} dx = \frac{\Gamma(69)}{2^{69}}.$$

SOLUTION-069

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{69} e^{-2x} dx = \frac{\Gamma(70)}{2^{70}}.$$

SOLUTION-070

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{70} e^{-2x} dx = \frac{\Gamma(71)}{2^{71}}.$$

SOLUTION-071

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{71} e^{-2x} dx = \frac{\Gamma(72)}{2^{72}}.$$

SOLUTION-072

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{72} e^{-2x} dx = \frac{\Gamma(73)}{2^{73}}.$$

SOLUTION-073

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{73} e^{-2x} dx = \frac{\Gamma(74)}{2^{74}}.$$

SOLUTION-074

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{74} e^{-2x} dx = \frac{\Gamma(75)}{2^{75}}.$$

SOLUTION-075

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{75} e^{-2x} dx = \frac{\Gamma(76)}{2^{76}}.$$

SOLUTION-076

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{76} e^{-2x} dx = \frac{\Gamma(77)}{2^{77}}.$$

SOLUTION-077

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{77} e^{-2x} dx = \frac{\Gamma(78)}{2^{78}}.$$

SOLUTION-078

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{78} e^{-2x} dx = \frac{\Gamma(79)}{2^{79}}.$$

SOLUTION-079

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{79} e^{-2x} dx = \frac{\Gamma(80)}{2^{80}}.$$

SOLUTION-080

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{80} e^{-2x} dx = \frac{\Gamma(81)}{2^{81}}.$$

SOLUTION-081

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{81} e^{-2x} dx = \frac{\Gamma(82)}{2^{82}}.$$

SOLUTION-082

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{82} e^{-2x} dx = \frac{\Gamma(83)}{2^{83}}.$$

SOLUTION-083

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{83} e^{-2x} dx = \frac{\Gamma(84)}{2^{84}}.$$

SOLUTION-084

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{84} e^{-2x} dx = \frac{\Gamma(85)}{2^{85}}.$$

SOLUTION-085

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{85} e^{-2x} dx = \frac{\Gamma(86)}{2^{86}}.$$

SOLUTION-086

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{86} e^{-2x} dx = \frac{\Gamma(87)}{2^{87}}.$$

SOLUTION-087

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{87} e^{-2x} dx = \frac{\Gamma(88)}{2^{88}}.$$

SOLUTION-088

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{88} e^{-2x} dx = \frac{\Gamma(89)}{2^{89}}.$$

SOLUTION-089

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{89} e^{-2x} dx = \frac{\Gamma(90)}{2^{90}}.$$

SOLUTION-090

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{90} e^{-2x} dx = \frac{\Gamma(91)}{2^{91}}.$$

SOLUTION-091

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{91} e^{-2x} dx = \frac{\Gamma(92)}{2^{92}}.$$

SOLUTION-092

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{92} e^{-2x} dx = \frac{\Gamma(93)}{2^{93}}.$$

SOLUTION-093

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{93} e^{-2x} dx = \frac{\Gamma(94)}{2^{94}}.$$

SOLUTION-094

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{94} e^{-2x} dx = \frac{\Gamma(95)}{2^{95}}.$$

SOLUTION-095

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{95} e^{-2x} dx = \frac{\Gamma(96)}{2^{96}}.$$

SOLUTION-096

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{96} e^{-2x} dx = \frac{\Gamma(97)}{2^{97}}.$$

SOLUTION-097

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{97} e^{-2x} dx = \frac{\Gamma(98)}{2^{98}}.$$

SOLUTION-098

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{98} e^{-2x} dx = \frac{\Gamma(99)}{2^{99}}.$$

SOLUTION-099

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{99} e^{-2x} dx = \frac{\Gamma(100)}{2^{100}}.$$

SOLUTION-100

Using the Gamma integral with the substitution $t = 2x$,

$$\int_0^{\infty} x^{100} e^{-2x} dx = \frac{\Gamma(101)}{2^{101}}.$$

INTEGRALS-DOCUMENT-END